BioSpecInfo

Software Bill of Materials (SBOM)

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Versione **bsi-v228** · commit `496b4ee`

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Software Bill of Materials (SBOM)

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Application version	`bsi-v228`
Commit	`496b4eefc761703f84cb8f72e9b5cd3d4dad6e0b`
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Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) _{RDKit / Greg Landrum}	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) _{RDKit / Greg Landrum}	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js _{mrdoob e collaboratori}	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass _{mrdoob e collaboratori}	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader _{mrdoob e collaboratori}	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js _{University of Pittsburgh / David Koes}	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer _{Daniel Probst}	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) _{Mozilla Foundation}	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker _{Mozilla Foundation}	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) _{sql.js / SQLite}	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) _{SQLite Consortium}	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) _{CERN Open Data Portal}	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.

Digests

File	SHA-256
`RDKit_minimal.js`	`58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9`
`RDKit_minimal.wasm`	`e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8`
`three.min.js`	`9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2`
`three_bloom.js`	`42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090`
`gltf_loader.js`	`5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6`
`3Dmol-min.js`	`06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff`
`smiles-drawer.min.js`	`955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2`
`vendor/pdfjs/pdf.min.mjs`	`bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259`
`vendor/pdfjs/pdf.worker.min.mjs`	`145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6`
`lib/sql-wasm.js`	`77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f`
`lib/sql-wasm.wasm`	`438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed`
`lib/dimuon.js`	`61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d`

2. Code written for this project

File	Size	Role
`index.html`	4676 kB	Main application: interface, teaching sections, chemical data.
`bsi-ai-hub.js`	422 kB	«Spectra» AI agent: agentic loop, tools, provider management.
`bsi-spettri.js`	35 kB	IR/NMR spectral prediction engine over the molecular graph.
`bsi-cheminfo.js`	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
`bsi-pretsch.js`	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
`bsi-nmr.js`	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
`bsi-geom3d.js`	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.
`bsi-fisica-comune.js`	15 kB	The shared home of the physics sections: header, tabs, sliders, canvas charts, language.
`bsi-fis-schrodinger.js`	22 kB	1D Schrödinger equation: finite differences, Sturm bisection, inverse iteration, Crank-Nicolson, Morse for real molecules.
`bsi-fis-hartreefock.js`	29 kB	Restricted Hartree-Fock in the STO-3G basis: McMurchie-Davidson integrals, Boys function, SCF with DIIS, molecular orbitals.
`bsi-fis-rotovib.js`	16 kB	IR and Raman rovibrational spectra of diatomics from molecular constants; combination differences.
`bsi-fis-termostat.js`	11 kB	Statistical thermodynamics: partition functions, entropy, heat capacity, Boltzmann populations.
`bsi-fis-dinamica.js`	12 kB	2D Lennard-Jones molecular dynamics: velocity Verlet, thermostat, virial, radial distribution.
`bsi-fis-matematica.js`	24 kB	Mathematics for chemistry: FFT of the FID, Hückel, Runge-Kutta for kinetics, least squares with errors.
`bsi-fis-cristalli.js`	20 kB	Crystallography: Bravais lattices, Miller indices, powder pattern with structure factor and absences.
`bsi-fis-nucleare.js`	14 kB	Radiochemistry: Bateman equations, radiometric dating, Weizsäcker semi-empirical formula.
`bsi-mat-cas.js`	53 kB	Computer-algebra engine: parser, exact fractions, canonical form, derivatives, verified antiderivatives, truncated series, limits.
`bsi-mat-basi.js`	26 kB	Mathematics basics: significant figures, logarithms and pH, equations with working, practice.

File	Size	Role
`bsi-mat-funzioni.js`	17 kB	Function analysis: domain, zeros, extrema, inflections, asymptotes; families and transformations.
`bsi-mat-calcolo.js`	21 kB	Derivatives with rules, integrals and Riemann sums, Taylor polynomials, limits.
`bsi-mat-lineare.js`	38 kB	Exact-fraction linear algebra: step-by-step Gauss-Jordan, determinant, inverse, eigenvalues, balancing as a null space.
`bsi-mat-ode.js`	21 kB	Differential equations: direction field, Euler/Heun/RK4, phase plane and equilibria from the Jacobian.
`bsi-mat-multi.js`	22 kB	Functions of several variables: surfaces, critical points from the Hessian, Maxwell relations, exact differentials.
`bsi-mat-complessi.js`	20 kB	Complex numbers (Argand plane, roots, Euler) and Fourier series.
`bsi-mat-statistica.js`	27 kB	Laboratory statistics: t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD and LOQ.
`bsi-nmr2d.js`	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
`bsi-moldraw.js`	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
`bsi-documento.js`	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
`bsi-quesito.js`	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
`bsi-spettrolettore.js`	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
`bsi-mol3d.js`	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
`bsi-lingue.js`	182 kB	Internationalisation engine: fourteen languages, writing direction included.
`bsi-molingue.js`	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
`bsi-pannelli-lingua.js`	66 kB	Panels for languages and chemical languages.
`sw.js`	6 kB	Service Worker: offline operation and network strategy.
`rdkit_lab.html`	260 kB	Cheminformatics laboratory.
`astro.html`	2660 kB	Astrochemistry module.
`chimorga.html`	194 kB	Organic chemistry module.
`proxy/spectra-proxy.js`	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
index.html	ab2fb6e0c0f0449d01211cfb969f45babc5ec57ae8bc4f4e79841f30bdf5fd55
bsi-ai-hub.js	985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476
bsi-spettri.js	be757f15a70f884e3374d401596e2dfd18bcc852cafbb4584920e2b2d24eb3c9
bsi-cheminfo.js	59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc
bsi-pretsch.js	27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3
bsi-nmr.js	881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72
bsi-geom3d.js	a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054
bsi-fisica-comune.js	092cde6b44f779694b0aa865f407a2e011a6d047878bf8e2fa3c819d6f3c23e0
bsi-fis-schrodinger.js	b032084511102c7231bcb244b5bb647128a080a18e9e6d706ec4c727ac29d17d
bsi-fis-hartreefock.js	8210d2a1501fe6e3f57c0ce8a0e6d5d78432274792833d50bc2e821fc7566fde
bsi-fis-rotovib.js	428182687a76111f013986127ee49beb7393498387b146bd0694ff16bea7d909
bsi-fis-termostat.js	0bb9f04d7d5b144791764fa4efd9f8e658be5b5d78b40f94e18752c2366589a9
bsi-fis-dinamica.js	f1d76c737ce7f75cdc9c75665194180a4cdf401e47a105583838f31c85c9b906
bsi-fis-matematica.js	616a2d4f338fe1d371f5ba3fd4fe569e94b2d94626a8d323bba8ae58f8550f62
bsi-fis-cristalli.js	f5283d63ff075ddc19f9d200194528366f38294909db5872e7604b46e0a7d7d5
bsi-fis-nucleare.js	8ae148a2f37a40f6902faa4148e44468b4d659b6c7d1230378de92c2703a5874
bsi-mat-cas.js	6e222b832095ff75f69db7939875d868a2b19da60a4b201a14d249ed834a7f52
bsi-mat-basi.js	db30e16c58bde3638f13de5c3311a00598628676b7b418dc023b175efa77c214
bsi-mat-funzioni.js	68f8780d4fb4fcda6f028e4749adef333253efbf270d0caab47def5bdf30c66c
bsi-mat-calcolo.js	c3363448c4d773d29c4f680eddb79e9ad55a5eb87b033a728754df8cbee2a631
bsi-mat-lineare.js	da33ea18f2b14c21c6e630be4a1ccfe2e35e911489bdd4ededd60c0f533ac94f
bsi-mat-ode.js	698725d49dc2bf862161914e41d38c3fafd33069021eae3b11b4973c146b52ce
bsi-mat-multi.js	1040861c3b9d340febe8c4a78d212b168eb42752ee22589facf8566a401234d7
bsi-mat-complessi.js	52a2ac5594a697d3a008d9fd629bee60fcff96ea76aa16c2ccda63eafd43f316
bsi-mat-statistica.js	0b0c82afa13173b84308589cf16fbaa8b30c5e575fd8fdee00034b70c34d17fc
bsi-nmr2d.js	a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78
bsi-moldraw.js	45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e
bsi-documento.js	0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9
bsi-quesito.js	6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf
bsi-spettrolettore.js	14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256
bsi-mol3d.js	93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893
bsi-lingue.js	d05cd5d99048f73f2985d271a833ee46c3076afca67cbd5390e1a7dedc23f26b
bsi-molingue.js	e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb

File	SHA-256
`bsi-pannelli-lingua.js`	`4af94d70ef104cf21be18ff7fbc7a2248c56e376e2ae48e8cac83878f4554f93`
`sw.js`	`11ee4642511db40e5e11461d7009f20e3cec4f30a429fb01a96a5629352d9292`
`rdkit_lab.html`	`066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad`
`astro.html`	`3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d`
`chimorga.html`	`047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d`
`proxy/spectra-proxy.js`	`808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962`

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
`playwright-core`	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

> **The application's licence is distinct from those of its components.** The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see `LICENSE`): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

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